



Show that the pre-typed term

$$\lambda x : ((\alpha \rightarrow \beta) \rightarrow \alpha). x (\lambda z : \alpha. y)$$

is legal: find a context Γ and a type ρ such that $\Gamma \vdash \lambda x : ((\alpha \rightarrow \beta) \rightarrow \alpha). x (\lambda z : \alpha. y) : \rho$.

- (a) Give the derivation in tree format.
 (b) Rewrite the same derivation in flag format.

Nederpelt & Geuvers, *Type Theory and Formal Proof* (CUP, 2014), Exercise 2.6.

I Definitions

Derivation rules of $\lambda \rightarrow$ (Def. 2.4.5).

$$\frac{}{\Gamma \vdash x : \sigma} \text{ (var) if } x : \sigma \in \Gamma \quad \frac{\Gamma \vdash M : \sigma \rightarrow \tau \quad \Gamma \vdash N : \sigma}{\Gamma \vdash MN : \tau} \text{ (appl)} \quad \frac{\Gamma, x : \sigma \vdash M : \tau}{\Gamma \vdash \lambda x : \sigma. M : \sigma \rightarrow \tau} \text{ (abst)}$$

Legal (Def. 2.4.10). M is legal if $\Gamma \vdash M : \rho$ for some context Γ and type ρ . A context declares the free variables of M ; each free variable needs a declaration, and the (var) rule can only use what the context declares.

Flag format (§2.5). Each declaration is written once, in a flag; the flag pole marks the lines it governs. Closing a flag is an (abst) step. Every line names the rule and the lines or flags it uses.

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Answer

Out of room? Continue on the back of the page.